

GCL-Detox: Disentangling Uniformity for Robust and Effective Graph Contrastive Learning-Based Recommendation

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Abstract

Graph Contrastive Learning (GCL) enhances recommendation via embedding uniformity, yet this very property introduces a critical vulnerability exploitable by poisoning attacks. Existing defenses overlook this geometric weakness. In this paper, we propose **GCL-Detox**, a defense framework that mitigates such attacks by systematically regulating embedding uniformity. Our key idea is a gradient-decoupled, tri-phase training strategy with progressive loss annealing, which enables a controlled transition from uniformity-driven representation learning to clustering-oriented recommendation optimization. We provide a rigorous theoretical analysis showing that the proposed training dynamics guarantee a monotonic increase in embedding non-uniformity, establish strict superiority over conventional joint training, and yield an explicit robustness bound against poisoning attacks. Extensive experiments on three real-world datasets confirm that GCL-Detox significantly enhances robustness against state-of-the-art attacks while maintaining competitive accuracy, validating both our theoretical and empirical claims.

CCS Concepts

• Information systems → Recommender systems.

Keywords

Graph Contrastive Learning, Recommender Systems, Poisoning Attacks, Robustness

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1 Introduction

Graph Contrastive Learning (GCL) has become a cornerstone of modern recommender systems [14, 26, 32–34] due to its ability to learn expressive representations from sparse interactions. By jointly optimizing self-supervised contrastive objectives (e.g., InfoNCE [13]) together with task-specific ranking losses such as Bayesian Personalized Ranking (BPR) [15], GCL-based recommenders effectively prevent representation collapse and enhance feature discriminability by enforcing embedding uniformity on the unit hypersphere [18–20, 22]. However, recent findings expose a critical security paradox [21, 22]: the very geometric property that ensures generalization also creates a latent attack surface. Adversaries can strategically exploit this predictable uniformity to amplify poisoning attacks [7, 8, 12, 16, 39]. In such attacks, malicious users inject carefully crafted interactions to artificially manipulate recommendation outcomes, promoting certain low-quality or unpopular items for profit.

To mitigate poisoning attacks in recommender systems, existing research has explored various defense strategies, which can be broadly categorized into three methodological strands: anomaly detection of malicious users [9, 40], trust-aware modeling [11, 30], and adversarial training techniques [1, 24, 27, 36]. While these approaches provide valuable insights and partial protection, they often operate at the data or optimization level and rely on explicit assumptions about attack behaviors. Moreover, many of them introduce additional training complexity or incur a non-negligible trade-off between robustness and recommendation accuracy. More fundamentally, most existing studies overlook a critical architectural factor inherent in GCL itself—the embedding uniformity induced by contrastive learning. Consequently, even robustly trained models may remain structurally susceptible to uniformity-exploiting attacks. Developing a defense that directly addresses this uniformity-induced vulnerability, without compromising recommendation quality, therefore remains a salient and unresolved problem.

In this paper, we propose **GCL-Detox**, a novel training framework that enhances the robustness of GCL-based recommender systems by *disentangling the role of embedding uniformity across*

different training stages. Instead of suppressing uniformity entirely, GCL-Detox revisits the conventional joint optimization paradigm and temporally decouples contrastive representation learning from recommendation optimization. Specifically, we introduce a gradient-decoupled *tri-phase training strategy*: (1) **Contrastive Pretraining**, in which the model is optimized solely with the contrastive loss to establish a well-initialized and semantically meaningful embedding space; (2) **Progressive Transition**, in which the contrastive loss is linearly annealed while the BPR loss is progressively emphasized; and (3) **BPR Fine-tuning**, relying exclusively on the ranking objective, refines accuracy while reducing the uniformity-exploitable attack surface.

Beyond empirical effectiveness, we establish a rigorous theoretical foundation for GCL-Detox. We prove that the proposed phased training with linear annealing guarantees a quantifiable and monotonic increase in embedding non-uniformity, accompanied by a provable lower bound. Furthermore, we formally demonstrate the strict superiority of GCL-Detox over conventional joint training, which is inherently constrained by a fixed trade-off between contrastive and ranking objectives. Building on this analysis, we derive an explicit theoretical bound on the reduction of attack effectiveness, offering a direct and testable connection between embedding geometry and robustness. Extensive experiments on multiple real-world datasets confirm that GCL-Detox significantly enhances robustness against state-of-the-art poisoning attacks, while maintaining or even improving recommendation accuracy.

Our contributions can be summarized as follows:

- We propose **GCL-Detox**, a novel defense framework for GCL-based recommender systems that enhances robustness against poisoning attacks by regulating embedding uniformity through a gradient-decoupled tri-phase training strategy.
- We establish a rigorous theoretical foundation for GCL-Detox, proving guaranteed improvements in embedding non-uniformity, strict superiority over joint training, and a quantitative robustness bound.
- Extensive experiments demonstrate that GCL-Detox significantly improves robustness against state-of-the-art poisoning attacks while maintaining or improving recommendation accuracy.

2 Preliminaries

2.1 GCL-based Recommendation

Let $\mathcal{D}$ denote the user–item interaction dataset, formulated as a bipartite graph $\mathcal{G} = (\mathcal{U} \cup \mathcal{I}, \mathcal{E})$, where $\mathcal{U}$, $\mathcal{I}$, and $\mathcal{E} \subseteq \mathcal{U} \times \mathcal{I}$ represent the user set, item set, and observed interactions, respectively. Graph-based recommender systems learn d -dimensional embeddings $\mathbf{E} = \{\mathbf{e}_v \in \mathbb{R}^d \mid v \in \mathcal{U} \cup \mathcal{I}\}$ using graph convolutional networks, typically LightGCN [5]. User preference is commonly predicted by the inner product $\mathbf{e}_u^\top \mathbf{e}_i$, and model training minimizes the Bayesian Personalized Ranking (BPR) loss [15, 21]:

$$\mathcal{L}_{\text{BPR}} = -\mathbb{E}_{(u,i,j) \sim \mathcal{P}_{\mathcal{D}}} \log \sigma(\mathbf{e}_u^\top \mathbf{e}_i - \mathbf{e}_u^\top \mathbf{e}_j), \quad (1)$$

where (u, i) and (u, j) denote positive and negative interactions, respectively.

Graph Contrastive Learning (GCL) augments graph recommendation by enforcing representation consistency across different augmented views of the same node. This objective is commonly

instantiated by the InfoNCE loss [13]:

$$\mathcal{L}_{\text{CL}} = - \sum_{p \in \mathcal{U} \cup \mathcal{I}} \log \frac{\exp(\mathbf{e}_{p'}^\top \mathbf{e}_{p''} / \tau)}{\sum_{v \in \mathcal{U} \cup \mathcal{I}} \exp(\mathbf{e}_{p'}^\top \mathbf{e}_v / \tau)}, \quad (2)$$

where p' and p'' are two augmented views of node p , and τ is a temperature parameter. Minimizing $\mathcal{L}_{\text{CL}}$ implicitly encourages embeddings to be uniformly distributed on the unit hypersphere, a geometric property that improves generalization but also plays a central role in this work.

Most existing GCL-based recommenders jointly optimize the recommendation and contrastive objectives:

$$\mathcal{L} = \mathcal{L}_{\text{BPR}} + \lambda \mathcal{L}_{\text{CL}}, \quad (3)$$

where λ balances task performance and representation regularization. While effective in practice, this joint optimization couples two objectives with inherently different geometric effects, which motivates the decoupled training strategy proposed in this paper.

2.2 Threat Model

- **Attacker.** The attacker aims to maximize the exposure of a target item $i \in \mathcal{I}$ by injecting a set of fake users $\mathcal{U}^P$ with carefully crafted interactions. We assume a white-box setting where the adversary has full knowledge of the model architecture and loss function, but operates under a bounded budget on the number of fake users and their interactions.
- **Defender.** The defender trains a recommendation model on potentially poisoned data without any prior knowledge of which users are fake, which items are targeted, or what attack strategy was employed. The defender's objective is twofold: preserve recommendation accuracy for genuine users while suppressing the artificial promotion of maliciously boosted items.

3 Methodology

We propose **GCL-Detox**, a training framework that enhances the robustness of GCL-based recommenders against poisoning attacks by *disentangling the role of embedding uniformity across different training stages*. Importantly, we do not argue that uniformity is harmful per se; rather, we show that *enforcing uniformity throughout the entire training process* can create a persistent geometric attack surface exploitable by adversaries. Our key idea is to *temporally decouple* contrastive representation learning from ranking optimization, thereby avoiding the simultaneous gradient conflict inherent in conventional joint training.

Tri-phase objective. The overall training process is governed by a piecewise objective controlled by the current epoch number t :

$$\mathcal{L}_{\text{detox}} = \begin{cases} \mathcal{L}_{\text{CL}}, & 0 < t \leq T_1, \\ \lambda(t) \mathcal{L}_{\text{CL}} + (1 - \lambda(t)) \mathcal{L}_{\text{BPR}}, & T_1 < t \leq T_2, \\ \mathcal{L}_{\text{BPR}}, & t > T_2, \end{cases} \quad (4)$$

where T_1 and T_2 mark the ends of Phase 1 and Phase 2, respectively. The function $\lambda(t)$ controls the influence of contrastive learning in the transition phase. Compared with warm-up or λ -tuning in joint training, our design is fundamentally different: *joint training retains simultaneous optimization and thus persistent gradient interference,*

whereas GCL-Detox *eliminates* the contrastive gradient in late training by annealing $\lambda(t)$ to zero, allowing the embedding geometry to move away from uniformity-dominated configurations.

3.1 Contrastive Pretraining Phase

Contrastive learning encourages a uniformly distributed embedding space, which provides a strong geometric prior and stabilizes optimization in sparse recommendation settings. Directly introducing the BPR objective in early epochs may prematurely align sparse (and potentially poisoned) interactions with user preferences, thereby absorbing noise and increasing susceptibility to attacks. Therefore, we perform pure representation learning in the first phase (epochs 1 to T_1):

$$\mathcal{L}_{\text{detox}}^{(1)} = \mathcal{L}_{\text{CL}}. \quad (5)$$

This phase serves two purposes:

- (1) **Stability and denoising.** It stabilizes early training by focusing solely on self-supervised contrastive signals, mitigating noise bias under sparsity.
- (2) **Geometry initialization.** It yields a well-initialized embedding space with strong uniformity, providing a robust starting point for subsequent ranking optimization.

3.2 Progressive Transition Phase

A direct switch from contrastive pretraining to pure BPR fine-tuning can cause abrupt geometric shifts, harming both optimization stability and robustness. To enable a smooth transition, we introduce an intermediate phase spanning epochs T_1+1 to T_2 , where the two losses are optimized with dynamically adjusted weights:

$$\mathcal{L}_{\text{detox}}^{(2)} = \lambda(t) \mathcal{L}_{\text{CL}} + (1 - \lambda(t)) \mathcal{L}_{\text{BPR}}, \quad t \in (T_1, T_2]. \quad (6)$$

We adopt a linear annealing schedule:

$$\lambda(t) = 1 - \frac{t - T_1}{T_2 - T_1}. \quad (7)$$

This design gradually shifts optimization from uniformity-driven representation learning to ranking-driven clustering, while avoiding abrupt “optimization shocks”. Crucially, because $(1 - \lambda(t))$ increases monotonically, the BPR gradient progressively comes to dominate and reshapes embeddings toward clustered configurations, thereby reducing the uniformity-exploitable attack surface. In Section 4, we show this linear annealing scheme yields a *monotonic and quantifiable* increase in embedding non-uniformity.

3.3 BPR Fine-tuning Phase

In the final phase ($t > T_2$), we remove the contrastive signal entirely and optimize solely with the BPR loss:

$$\mathcal{L}_{\text{detox}}^{(3)} = \mathcal{L}_{\text{BPR}}. \quad (8)$$

This phase focuses on:

- (1) **Utility refinement.** It refines recommendation performance by fully leveraging supervised ranking signals.
- (2) **Attack surface reduction.** It further reduces susceptibility to uniformity-exploiting poisoning attacks by eliminating the uniformity-enforcing gradient constraint from contrastive learning.

Algorithm 1 in Appendix B.1 illustrates the training process of the tri-phase framework. It shows that we can achieve a more robust recommendation model with almost no additional computational overhead on the basis of the original graph contrastive learning-based recommendation system. This demonstrates the simplicity and efficiency of the proposed GCL-Detox method, as well as its easy integration into other models.

4 Theoretical Analysis

Conventional joint training suffers from persistent gradient conflict: $\mathcal{L}_{\text{CL}}$ enforces uniformity while $\mathcal{L}_{\text{BPR}}$ drives clustering, yielding a brittle equilibrium exploitable by poisoning attacks. GCL-Detox resolves this via temporal decoupling and contrastive annealing, enabling a smooth transition from uniformity to clustered geometries. We first introduce formal definitions and assumptions, then present two theorems: Theorem 4.1 guarantees monotonic non-uniformity increase, and Theorem 4.2 establishes strict superiority over joint training. Corollary 4.1 links these gains to provable robustness improvements.

4.1 Basic Definitions and Assumptions

Recall that $\mathbf{E} \in \mathbb{R}^{|\mathcal{U} \cup \mathcal{I}| \times d}$ is the embedding matrix. We consider normalized embeddings: $\|\mathbf{e}_i\|_2 = 1$ for all $i \in \{1, \dots, |\mathcal{U} \cup \mathcal{I}|\}$. Let $\mathcal{L}_{\text{CL}}(\mathbf{E})$ denote the contrastive loss and $\mathcal{L}_{\text{BPR}}(\Theta)$ the BPR loss, where $\Theta = (\mathbf{E}, \Phi)$ includes embeddings and ranking-network parameters.

Non-uniformity measure. We adopt the uniformity-based measure widely used in prior work [20, 22]:

$$\psi(\mathbf{E}) = \log \mathbb{E}_{i,j \sim P_{\mathcal{D}}} \left[\exp(-\tau \|\mathbf{e}_i - \mathbf{e}_j\|_2^2 / 2) \right], \quad (9)$$

where $\tau > 0$ is a temperature parameter. Larger $\psi(\mathbf{E})$ indicates higher *non-uniformity* (i.e., more clustered embeddings), while smaller values indicate more uniform distributions on the unit hypersphere. We denote $\psi_{\min} = \min_{\mathbf{E}} \psi(\mathbf{E})$ as the minimal non-uniformity achievable under contrastive learning.

We introduce standard smoothness and alignment assumptions commonly adopted in nonconvex optimization and contrastive learning analysis. Specifically, we assume: (A1) Smoothness of Non-uniformity; (A2) Gradient Alignment at Transition Start; (A3) Bounded Ranking Gradient; and (A4) Approximate Stationarity after Contrastive Pretraining. Full formal statements are provided in Appendix C.

4.2 Non-uniformity Guarantee of GCL-Detox

THEOREM 4.1 (NON-UNIFORMITY INCREASE VIA LINEAR ANNEALING). *Under Assumptions (A1)–(A4), consider the progressive transition phase with $\lambda(t) = 1 - \frac{t - T_1}{T_2 - T_1}$ for $t \in (T_1, T_2]$. Let E_t denote the embeddings at epoch t . Then for any $t \in (T_1, T_2]$, the non-uniformity increase satisfies*

$$\psi(E_t) - \psi(E_{T_1}) \geq (t - T_1) \eta \left(\alpha - C_1 \eta (t - T_1) \right) - O(\eta^3) - O(\eta \varepsilon),$$

where η is the learning rate and $C_1 > 0$ is a constant. For sufficiently small η and ε , $\psi(E_t)$ increases monotonically with t .

PROOF SKETCH. At epoch t , the embedding update uses the convex-combination gradient $\nabla_{\mathbf{E}} \mathcal{L}_{\text{total}}(t) = \lambda(t) \nabla \mathcal{L}_{\text{CL}} + (1 - \lambda(t)) \nabla \mathcal{L}_{\text{BPR}}$.

Since Phase 1 reaches an ε -stationary point of $\mathcal{L}_{\text{CL}}$, the contribution of $\nabla \mathcal{L}_{\text{CL}}$ during the transition is bounded and produces at most a $-\mathcal{O}(\eta\varepsilon)$ term. By L_ψ -smoothness of ψ and the alignment in (A2), one step yields a gain of order $(1 - \lambda(t))\eta\alpha$ minus a quadratic smoothness penalty of order $L_\psi\eta^2(1 - \lambda(t))^2G^2$. Summing over t and substituting the linear form of $\lambda(t)$ gives the stated bound. $\square$

Remark. Theorem 4.1 explains why linear annealing is effective: the accumulated gain is proportional to the integral of $(1 - \lambda(\tau))$ over epochs, yielding a predictable increase of $\psi(\mathbf{E}_t)$. Other schedules can be used, but linear annealing offers both stability and analytical tractability.

4.3 Superiority over Joint Training

While Theorem 4.1 shows that staged training guarantees a monotonic increase in embedding non-uniformity, we now demonstrate that such an improvement cannot be fully achieved by joint optimization.

Joint training minimizes

$$\mathcal{L}_{\text{JT}} = \lambda_{\text{JT}}\mathcal{L}_{\text{CL}} + (1 - \lambda_{\text{JT}})\mathcal{L}_{\text{BPR}}, \quad \lambda_{\text{JT}} \in (0, 1).$$

At convergence, the contrastive and ranking gradients must balance each other. Since contrastive learning enforces embedding uniformity, any fixed $\lambda_{\text{JT}} > 0$ induces a persistent *uniformity-restoring force*, which limits the achievable embedding non-uniformity. In contrast, GCL-Detox linearly anneals the contrastive weight $\lambda(t)$ to zero, completely removing the uniformity-restoring constraint. This allows the ranking objective to dominate the optimization, yielding a strictly higher non-uniformity as formally established below.

THEOREM 4.2 (STRICT SUPERIORITY OVER JOINT TRAINING). *Let $\mathbf{E}^{\text{TS}}$ be the final embedding learned by GCL-Detox (tri-phase staged training), and let $\mathbf{E}^{\text{JT}}$ be the final embedding learned by joint training that minimizes $\mathcal{L}_{\text{JT}} = \lambda_{\text{JT}}\mathcal{L}_{\text{CL}} + (1 - \lambda_{\text{JT}})\mathcal{L}_{\text{BPR}}$ with a fixed $\lambda_{\text{JT}} \in (0, 1)$. Under the conditions of Theorem 4.1, there exist constants $C_2 > 0$ and $\delta = \mathcal{O}(\eta^3) + \mathcal{O}(\eta\varepsilon)$ such that*

$$\psi(\mathbf{E}^{\text{TS}}) - \psi(\mathbf{E}^{\text{JT}}) \geq \eta(T_2 - T_1) \left(\alpha - C_1\eta(T_2 - T_1) \right) - C_2 \left(\frac{1 - \lambda_{\text{JT}}}{\lambda_{\text{JT}}} \right)^2 - \delta. \quad (10)$$

Consequently, for sufficiently small η and ε , the staged training yields strictly higher non-uniformity: $\psi(\mathbf{E}^{\text{TS}}) > \psi(\mathbf{E}^{\text{JT}})$.

PROOF SKETCH. Joint training converges to a stationary point satisfying $\lambda_{\text{JT}}\nabla \mathcal{L}_{\text{CL}} + (1 - \lambda_{\text{JT}})\nabla \mathcal{L}_{\text{BPR}} = 0$, which enforces a persistent balance between contrastive and ranking gradients. Under standard smoothness and local regularity assumptions, this equilibrium imposes an upper bound on the achievable embedding non-uniformity as long as $\lambda_{\text{JT}} > 0$. In contrast, GCL-Detox removes the contrastive constraint by annealing $\lambda(t)$ to zero, enabling the guaranteed non-uniformity increase established in Theorem 4.1. Combining the staged lower bound with the joint-training upper bound yields the strict gap in Eq. (10). $\square$

Remark. Eq. (10) explains why tuning λ_{JT} in joint training cannot fully match GCL-Detox: any fixed $\lambda_{\text{JT}} > 0$ induces a persistent equilibrium constraint, whereas our staged design ultimately removes the contrastive constraint by setting $\lambda(t) = 0$.

4.4 Robustness Implications

COROLLARY 4.1 (THEORY-BACKED ROBUSTNESS GUARANTEE). *Our analysis suggests that increasing embedding non-uniformity ψ helps reduce poisoning attack effectiveness. If the training hyper-parameters satisfy the theory-backed tuning rule*

$$\eta(T_2 - T_1) < \frac{\alpha}{C_1}, \quad (11)$$

then $\Delta_\psi := \psi(\mathbf{E}^{\text{TS}}) - \psi(\mathbf{E}^{\text{JT}}) > 0$, and yielding improved robustness. That is, GCL-Detox strictly reduces the attack effectiveness compared to joint training under condition (11).

This theoretical guideline aligns with our empirical observations in Section 5.

4.5 Discussion

Our analysis provides the following insights:

- (1) **Why phased optimization is necessary.** Joint training with any fixed λ_{JT} induces a persistent equilibrium constraint that caps non-uniformity and preserves the geometric structure exploitable by uniformity-based attacks. GCL-Detox avoids this through temporal decoupling and the eventual removal of the contrastive signal.
- (2) **Why linear annealing works.** Theorem 4.1 shows that linear annealing yields a controlled, monotonic increase of $\psi(\mathbf{E}_t)$ during transition, enabling stable and predictable geometry evolution.
- (3) **How geometry implies robustness.** Corollary 4.1 links the non-uniformity gap to an explicit reduction in attack effectiveness, explaining why moderate increases in ψ can lead to substantial robustness gains.
- (4) **Practical tuning guidance.** The condition $\eta(T_2 - T_1) < \alpha/C_1$ provides a theory-grounded guideline for selecting the learning rate and transition length.

These insights align with our empirical results: $\psi(\mathbf{E})$ increases monotonically during the transition phase, and attack success decreases as ψ grows, supporting the robustness prediction implied by our theory. Ultimately, GCL-Detox’s strength stems from its *optimization dynamics*—a controlled transition from uniformity-dominated representations to task-specific clustering that reduces the geometric attack surface exploited by poisoning attacks.

5 Experiments

To comprehensively evaluate the effectiveness of GCL-Detox, we conduct experiments designed to address the following key research questions:

- RQ1** Does GCL-Detox effectively enhance robustness against poisoning attacks while maintaining competitive recommendation accuracy?
- RQ2** How does the embedding geometry evolve during training—in particular, how does the (non-)uniformity metric change across phases under strong attacks (e.g., CLear), and does it support the theoretical training dynamics predicted in Section 4?
- RQ3** How do the key design components—the tri-phase structure and the linear annealing schedule—contribute to the overall performance?

Dataset	#Users	#Items	#Interactions	Sparsity
DouBan	2,807	33,426	894,887	99.05%
ML-10M	68,117	9,181	2,129,834	99.66%
Yelp	31,668	38,048	1,561,406	99.87%

Table 1: Dataset statistics.

RQ4 How sensitive is GCL-Detox to its hyperparameters (T_1, T_2), and does practical tuning align with the theoretical guideline on the transition length (e.g., controlling $\eta(T_2 - T_1)$)?

RQ5 Does GCL-Detox effectively prevent the item-level ranking manipulation targeted by poisoning attacks, especially for unpopular items?

5.1 Experimental Settings

Datasets. The experiments are conducted on three public datasets with various scales: ML-10M [4], DouBan [38], and Yelp [34], which are summarized in Table 1.

GCL-based Recommendation Models. We choose LightGCN [5] as the backbone model, and combine it with the following three recent GCL objectives to form the GCL-based recommenders to defend: **SGL** [26], **SimGCL** [34], and **XSimGCL** [32].

Attack Methods. We utilize representative poisoning attacks: **Rand** [7], **PGA** [8], **A_ra** [16] and **CLear** [22], where CLear explicitly strengthens poisoning by intensifying embedding uniformity. We also report results under no attack (**NoAtt**).

Baseline Defense Methods. We compare GCL-Detox with **Degree-SAD** [9], **Anti-FU** [31], and **Trust-GRS** [11]. **Origin** denotes the undefended GCL-based recommender. For ablation, we include **w/o_p** (removing the contrastive pretraining phase) and **w/o_t** (removing the progressive transition phase).

Evaluation Protocol. Following [22], we split each dataset into training/validation/test with ratio 7:1:2. We evaluate recommendation accuracy using NDCG@50 on the clean test set, and robustness using HR@50, defined as the fraction of users whose top-50 lists contain the targeted items. The trade-off between utility and robustness is measured by

$$\text{Trade_off} = 0.5 \cdot \frac{\text{NDCG} - \text{NDCG}'}{\text{NDCG}'} + 0.5 \cdot \frac{\text{HR}' - \text{HR}}{\text{HR}'},$$

where NDCG' and HR' are results of **Origin**. Consistent with [11, 22], we randomly select five target items from the bottom 20% of the popularity ranking. Each experiment is repeated three times and mean results are reported.

5.2 Recommendation Performance and Robustness (RQ1)

Table 2 summarizes the utility–robustness results on DouBan, where the hyperparameters T_1 and $T_2 - T_1$ during training are determined by the optimal results shown in Figure 4 (i.e., T_1 and $T_2 - T_1$ corresponding to the red points). We highlight three key observations.

(1) **Best utility across victim models.** Across all three victim models, GCL-Detox achieves the best NDCG@50 under all attack settings. This indicates that our staged optimization not only preserves recommendation utility but also consistently yields more discriminative embeddings.

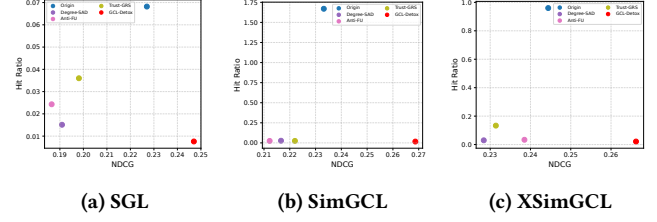


Figure 1: Pareto frontier of three victim models (a) SGL, (b) SimGCL, and (c) XSimGCL on DouBan under CLear.

- (2) **Robustness is strongest against uniformity-exploiting attacks (CLear and A_ra).** Under CLear, GCL-Detox achieves the lowest HR@50 for SimGCL, XSimGCL, and SGL, i.e., it most effectively suppresses attack-induced promotion. A similar pattern holds under the adaptive attack A_ra, where GCL-Detox again attains the lowest HR@50 for all three victim models. For Rand and PGA, some baselines may obtain a lower HR@50 in specific cases (e.g., SimGCL), but this typically comes with inferior utility and thus does not translate into the best overall trade-off.
- (3) **Best overall trade-off in 11/12 settings.** GCL-Detox achieves the highest Trade-off scores for all attacks on SimGCL and XSimGCL, and for PGA/A_ra/CLear on SGL. The only exception is SGL under Rand, where w/o_p is the best and GCL-Detox is the second best. Overall, these results demonstrate that GCL-Detox provides the most favorable utility–robustness balance on DouBan.

Similar conclusions can be drawn from ML-10M and Yelp (Tables 5 and 6 in Appendix B.3).

5.3 Visualization of the Trade-off (RQ1)

Figure 1 visualizes the utility–robustness Pareto frontier under CLear. GCL-Detox consistently lies in or near the desirable bottom-right region (high NDCG, low HR), showing that it reduces attack success rate while maintaining competitive utility. Baseline defenses exhibit inferior trade-offs across victim models. The trade-off results on the ML-10M and Yelp datasets are presented in Figures 6 and 7 in Appendix B.3, where our method is also shown to achieve the best performance in balancing robustness and recommendation accuracy.

5.4 Robustness Under Varying Attack Strengths (RQ1)

Figure 2 shows robustness under increasing poisoning budgets. While all methods degrade as the attack strengthens, GCL-Detox consistently achieves the lowest or near-lowest HR@50 across budgets and victim models, indicating stable protection against strong, uniformity-exploiting attacks. The results on ML-10M and Yelp are shown in Figures 8 and 9 in Appendix B.3, where similar conclusions are observed, where similar conclusions are observed.

Victim Model	Defense	NDCG@50 $\uparrow$					HR@50 ($\times 10^{-2}$) $\downarrow$					Trade-off $\uparrow$			
		NoAtt	Rand	PGA	A_ra	CLear	NoAtt	Rand	PGA	A_ra	CLear	Rand	PGA	A_ra	CLear
SimGCL	Origin	0.2264	0.2323	0.2267	0.2327	0.2332	0.0000	2.3317	1.7658	1.9603	1.6698	-	-	-	-
	Degree_SAD	0.2129	0.2178	0.2141	0.2176	0.2166	0.0000	0.0342	0.0435	0.0434	0.0268	0.4614	0.4599	0.4565	0.4564
	Anti-FU	0.2103	0.2129	0.2099	0.2141	0.2122	0.0000	0.0284	<u>0.0306</u>	0.0266	<u>0.0246</u>	0.4523	0.4542	0.4534	0.4476
	Trust-GRS	0.2256	0.2338	0.2314	0.2270	0.2220	0.0000	0.0270	0.0184	0.0232	0.0248	0.4975	0.5052	0.4820	0.4687
	w/o_p	<u>0.2629</u>	<u>0.2613</u>	<u>0.2620</u>	<u>0.2634</u>	<u>0.2594</u>	0.0000	<u>0.0051</u>	0.0505	<u>0.0227</u>	0.0480	<u>0.5615</u>	0.5636	<u>0.5604</u>	<u>0.5417</u>
	w/o_t	0.2617	0.2598	0.2606	0.2622	0.2577	0.0000	0.0025	0.0354	0.0253	0.0429	0.5587	0.5647	0.5571	0.5396
XSimGCL	GCL-Detox	0.2689	0.2681	0.2688	0.2684	0.2622	0.0000	0.0076	0.0493	0.0177	0.0164	0.5754	0.5789	0.5723	0.5573
	Origin	0.2399	0.2455	0.2411	0.2449	0.2451	0.0000	1.2631	1.3111	1.0888	0.9600	-	-	-	-
	Degree_SAD	0.2403	0.2381	0.2343	0.2392	0.2295	0.0000	0.0272	<u>0.0266</u>	<u>0.0254</u>	<u>0.0299</u>	0.4743	0.4757	0.4768	0.4526
	Anti-FU	0.2355	0.2411	0.2366	0.2392	0.2388	0.0000	0.0201	0.0403	0.0256	0.0337	0.4832	0.4752	0.4767	0.4697
	Trust-GRS	0.2267	0.2301	0.2359	0.2326	0.2341	0.0000	0.0948	0.0202	0.2666	0.1332	0.4312	0.4815	0.3526	0.4083
	w/o_p	0.2593	<u>0.2598</u>	<u>0.2595</u>	0.2594	<u>0.2726</u>	0.0000	0.0101	0.0581	0.0328	0.0454	<u>0.5252</u>	0.5160	<u>0.5146</u>	<u>0.5326</u>
SGL	w/o_t	0.2602	0.2584	0.2589	<u>0.2598</u>	0.2585	0.0000	<u>0.0076</u>	0.0505	0.0379	0.0420	0.5234	<u>0.5175</u>	0.5131	0.5055
	GCL-Detox	0.2666	0.2661	0.2655	0.2664	0.2736	0.0000	0.0025	0.0429	0.0177	0.0213	0.5410	0.5342	0.5359	0.5471
	Origin	0.2209	0.2251	0.2190	0.2281	0.2270	0.0000	0.2122	0.2703	0.0657	0.0682	-	-	-	-
	Degree_SAD	0.1863	0.1907	0.1854	0.1920	0.1909	0.0000	0.0116	0.0271	0.0179	<u>0.0151</u>	0.3961	0.3733	0.3073	0.3097
	Anti-FU	0.1844	0.1871	0.1846	0.1884	0.1865	0.0008	0.0122	0.0358	<u>0.0160</u>	0.0243	0.3868	0.3553	0.2916	0.2316
	Trust-GRS	0.1951	0.1929	0.1972	0.1998	0.1981	0.0000	0.0356	0.0203	0.0364	0.0360	0.3445	0.4127	0.1608	0.1728
SGL	w/o_p	0.2454	<u>0.2455</u>	<u>0.2466</u>	<u>0.2459</u>	<u>0.2540</u>	0.0000	0.0000	0.0025	0.0278	0.0421	0.5451	<u>0.5584</u>	0.3275	0.2511
	w/o_t	0.2467	0.2442	0.2457	0.2449	0.2459	0.0000	0.0051	0.0076	0.0215	0.0369	0.5304	0.5470	0.3735	0.2714
	GCL-Detox	0.2478	0.2472	0.2478	0.2485	0.2554	0.0000	<u>0.0051</u>	0.0025	0.0152	0.0076	<u>0.5370</u>	0.5611	0.4295	0.5069

Table 2: Comparison on DouBan dataset. Best results are in bold, and second best are underlined.

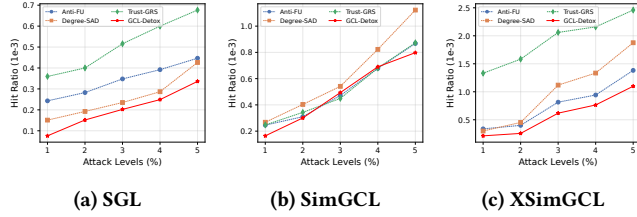
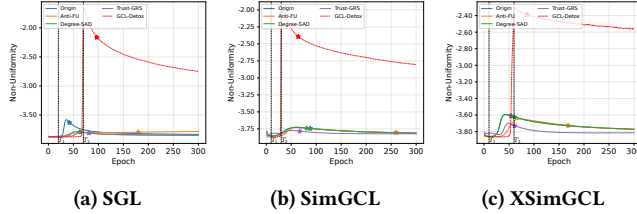
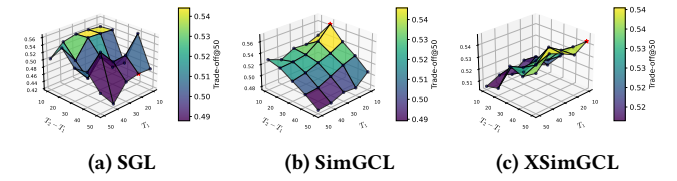


Figure 2: HR@50 under CLear with varying poisoning budgets (1%–5% fake users) on DouBan.

Figure 3: Evolution of the (non-)uniformity metric over epochs on DouBan under CLear. T_1 and T_2 indicate the phase boundaries of GCL-Detox.

5.5 Embedding (Non-)Uniformity Dynamics Under CLear (RQ2)

To validate the training dynamics predicted by our theory (Section 4), we visualize the evolution of the embedding (non-)uniformity metric during training under the strongest attack, CLear, on DouBan. Figure 3 reports curves for all defenses across three victim models, where T_1 and T_2 mark the phase boundaries of GCL-Detox. The

Figure 4: Trade-off@50 under CLear on DouBan when tuning (T_1, T_2) (Phase 1 end and Phase 2 end).

results on ML-10M and Yelp are shown in Figures 10 and 11 in Appendix B.3).

We observe a clear phase-wise behavior for GCL-Detox: (i) during contrastive pretraining ($t \leq T_1$), the geometry stays close to a uniformity-dominated regime; (ii) during the progressive transition ($T_1 < t \leq T_2$), the metric changes *monotonically* in the direction of reduced uniformity (i.e., increased clustering / higher non-uniformity), consistent with the monotonic increase guarantee established in Theorem 4.1; (iii) once entering BPR fine-tuning ($t > T_2$), the geometry stabilizes at a more clustered configuration. In contrast, joint training (Origin) and baseline defenses typically converge early to relatively stable geometry, suggesting that the persistent coupling between contrastive and ranking gradients may cap the achievable geometry shift, which aligns with Theorem 4.2. Overall, these results empirically support the core mechanism of GCL-Detox: *temporal decoupling with annealing* provides a controlled evolution from uniformity toward clustering, which in turn improves robustness against uniformity-exploiting attacks.

5.6 Ablation Study on Key Components (RQ3)

We conduct ablation studies with two variants: **w/o_p** removes contrastive pretraining, and **w/o_t** removes the progressive transition phase. As shown in Table 2, both variants underperform the full

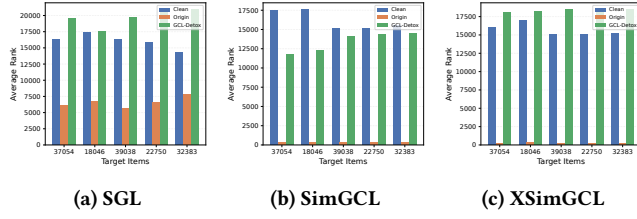


Figure 5: Average ranking positions of five unpopular target items under CLeaR on DouBan. Lower ranks indicate worse recommendation positions.

GCL-Detox in terms of the performance–robustness trade-off under strong attacks (notably CLeaR), demonstrating that the gains do not come from a single component. Removing pretraining weakens the initial embedding geometry and may harm stability under sparsity, while removing the transition causes an abrupt objective shift that can disrupt the evolution of embedding geometry. These results support our design rationale that *phased optimization and smooth annealing must work together* to achieve stable geometry reshaping and robust recommendation. Similar conclusions can be drawn from ML-10M and Yelp datasets (Tables 5 and 6 in Appendix B.3).

5.7 Hyperparameter Sensitivity and Theory-backed Tuning Rule (RQ4)

We analyze the sensitivity of GCL-Detox to the phase-boundary hyperparameters (T_1, T_2). Figure 4 reports Trade-off@50 heatmaps under CLeaR on DouBan. Overall, GCL-Detox remains effective across a broad range of (T_1, T_2), and strong performance typically occurs when (i) Phase 1 is long enough to provide a stable, uniformity-dominated initialization and (ii) the transition length ($T_2 - T_1$) is moderate, enabling a smooth geometry shift without abrupt optimization shocks. The tuning results for ML-10M and Yelp are shown in Figure 12 and 13 in Appendix B.3.

Theory-backed tuning rule. Beyond empirical tuning, our theory provides an explicit and testable guideline for selecting the transition length. Specifically, Corollary 4.1 states that the robustness improvement is guaranteed when the hyperparameters satisfy $\eta(T_2 - T_1) < \frac{\alpha}{C_1}$ (Eq. (11)), where η is the learning rate, α is the alignment constant in Assumption (A2), and $C_1 = \frac{L_\psi G^2}{2}$ is determined by the smoothness and gradient bound in Assumptions (A1) and (A3). This condition is *not* a heuristic: it directly enforces a positive non-uniformity gap $\Delta_\psi > 0$ in Theorem 4.2, which in turn implies a strict reduction of attack effectiveness.

Table 3 verifies Eq. (11) under our adopted hyperparameters across all datasets and victim models; all cases satisfy the condition. This provides a concrete, theory-grounded explanation for why GCL-Detox remains stable and consistently robust in our tuning results: the transition is sufficiently smooth (controlled by $\eta(T_2 - T_1)$) to guarantee monotonic non-uniformity improvement and a positive robustness gap. In practice, we recommend using Eq. (11) as a *principled* guideline to set (T_1, T_2): select T_1 to ensure adequate contrastive initialization, and choose a transition length ($T_2 - T_1$) that satisfies the inequality for the given learning rate η .

Dataset	Model	L_ψ	G	$\alpha (\times 10^{-7})$	η	$C_1 (\times 10^{-7})$	$T_2 - T_1$	Satisfying Eq. (11)?
DouBan	SimGCL	0.0014	0.0014	17.8208	0.0010	0.0730	20	✓
	XSimGCL	0.0165	0.0026	15.6334	0.0010	0.5660	50	✓
	SGL	0.0242	0.0031	2.9907	0.0010	1.2010	50	✓
ML-10M	SimGCL	0.0003	0.0015	0.6674	0.0010	0.0036	10	✓
	XSimGCL	0.0078	0.0011	4.7500	0.0010	0.0434	20	✓
	SGL	0.0015	0.0013	1.7277	0.0010	0.0132	10	✓
Yelp	SimGCL	0.0004	0.0005	0.1458	0.0010	0.0004	10	✓
	XSimGCL	0.0035	0.0004	0.0548	0.0010	0.0027	30	✓
	SGL	0.0010	0.0005	0.2054	0.0010	0.0010	10	✓

Table 3: Verification of the theory-backed tuning rule $\eta(T_2 - T_1) < \alpha/C_1$ across three datasets and three GCL-based models. L_ψ and G correspond to the smoothness and gradient-bound constants in Assumptions (A1) and (A3), respectively. All evaluated configurations satisfy the theoretical condition (TRUE), supporting the practical validity of Corollary 4.1.

5.8 Case Study: Item-level Ranking Behavior under CLeaR (RQ5)

To address RQ5, we conduct a case study to examine how GCL-Detox mitigates item-level ranking manipulation under poisoning attacks. On the DouBan dataset, we randomly sample five unpopular items (from the bottom 20% of the popularity ranking under clean training) as attack targets and analyze their average recommendation ranks under the CLeaR attack. Figure 5 shows that CLeaR effectively promotes these target items to substantially higher ranks compared to the clean baseline, demonstrating its ability to exploit the embedding geometry learned by GCL-based models. Under attack, both the undefended joint-training scheme (Origin) and GCL-Detox exhibit ranking elevation to varying degrees. However, across all three victim models, GCL-Detox consistently yields ranking positions that are closest to the clean results, indicating that it significantly suppresses attack-induced ranking deviation, rather than completely eliminating ranking shifts.

This case study complements our quantitative evaluation by providing direct item-level evidence that GCL-Detox effectively mitigates the core effect of poisoning attacks—artificial promotion of unpopular items—by preserving the original ranking geometry. Similar item-level ranking behaviors are consistently observed on the ML-10M and Yelp datasets, as shown in Figures 14 and 15 in Appendix B.3.

6 Conclusion

Embedding uniformity is a fundamental source of vulnerability in GCL-based recommender systems under poisoning attacks. Our defense, GCL-Detox, mitigates it via gradient-decoupled tri-phase training with progressive annealing, with theoretical guarantees on non-uniformity improvement, superiority over joint training, and robustness. Experiments across datasets and models show stronger robustness, preserved utility, and no attack-induced promotion of unpopular items.

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GenAI disclosure statement

ChatGPT-4o was solely used to check spelling and improve grammar during the writing of this paper.

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Algorithm 1 GCL-Detox: Unified Tri-Phase Training Framework

Require: User-item interaction graph $\mathcal{G}$, learning rate η , total epochs T_{total}

Hyperparameters: T_1, T_2 (phase thresholds), τ (temperature), d (embedding dim)

Parameter: User-item embedding matrix $\mathbf{E} \in \mathbb{R}^{(N+M) \times d}$

Ensure: Robust user/item embeddings $\mathbf{E}$

- 1: Initialize $\mathbf{E}$ randomly and normalize to unit hypersphere
- 2: **for** $t = 1$ **to** T_{total} **do**
- 3: Compute annealing coefficient:

$$\lambda(t) = \begin{cases} 1 & t \leq T_1, \\ 1 - \frac{t - T_1}{T_2 - T_1} & T_1 < t \leq T_2, \\ 0 & t > T_2 \end{cases}$$

- 4: Generate graph augmentations and compute contrastive loss $\mathcal{L}_{\text{CL}}$
- 5: Sample BPR triplets and compute ranking loss $\mathcal{L}_{\text{BPR}}$
- 6: Total loss: $\mathcal{L} = \lambda(t) \cdot \mathcal{L}_{\text{CL}} + (1 - \lambda(t)) \cdot \mathcal{L}_{\text{BPR}}$
- 7: Update embeddings: $\mathbf{E} \leftarrow \mathbf{E} - \eta \nabla_{\mathbf{E}} \mathcal{L}$
- 8: **end for**
- 9: **return** Robust embedding matrix $\mathbf{E}$

Result	DouBan			ML-10M			Yelp		
	SimGCL	XSimGCL	SGL	SimGCL	XSimGCL	SGL	SimGCL	XSimGCL	SGL
T_1	10	10	20	10	20	10	20	20	40
$T_2 - T_1$	20	50	50	20	10	10	10	30	10

Table 4: Hyperparameter settings under CLear.

A Related Work

Graph Contrastive Learning (GCL) has been widely adopted in recommender systems to alleviate data sparsity and cold-start issues by learning invariant representations from multiple graph views. Most GCL-based recommenders jointly optimize a contrastive objective (e.g., InfoNCE) with a recommendation loss (e.g., BPR). Representative methods include SGL [26], which constructs augmented views via node or edge dropout, and SimGCL [34] and XSimGCL [32], which introduce embedding-level perturbations and explicitly encourage embedding uniformity. Beyond joint training, some works adopt pretraining–finetuning or parameter-efficient tuning paradigms to reduce task interference and improve generalization [28, 29, 35, 37]. While these approaches achieve strong recommendation performance, they largely overlook the security implications of the uniform embedding geometry induced by contrastive learning.

Poisoning attacks pose a serious threat to recommender systems by injecting malicious users or interactions to manipulate recommendation outcomes. Early attacks rely on heuristic strategies such as random or average profile construction [7], while later works propose optimization-based or generative methods to craft more effective and stealthy attacks [3, 8, 10, 17, 23, 25]. Recently, CLear [22] reveals a critical vulnerability specific to GCL-based recommenders by explicitly exploiting the embedding uniformity enforced by contrastive objectives, significantly amplifying attack

effectiveness. This line of work highlights the tight connection between representation geometry and adversarial manipulation.

Defenses against poisoning attacks in recommender systems generally fall into data filtering and robust training. Data filtering methods aim to identify and remove suspicious users or interactions, such as Anti-FU [31], Degree-SAD [9], Trust-GRS [11], and De-Pois [2]. Robust training approaches instead enhance model resilience during optimization, for example through adversarial training or vulnerability-aware perturbations [6, 24, 36]. However, existing defenses do not explicitly address the embedding uniformity inherent in GCL-based recommenders, and thus remain limited against attacks that directly exploit this geometric property. In contrast, our work targets this overlooked vulnerability by regulating embedding uniformity through a staged, theory-grounded training strategy.

B More Experimental Settings and Results

B.1 Implementation Details of GCL-Detox Training Framework

Algorithm 1 illustrates the training process of the GCL-Detox framework.

B.2 Hyperparameter Tuning

In our experiments, T_1 and $T_2 - T_1$ are chosen from the search space $\{10, 20, 30, 40, 50\}$ and optimized via grid search. The hyperparameter tuning results on the three datasets under CLear are illustrated in Figures 4, 12 and 13, where the red dots correspond to the optimal hyperparameter configurations. We summarize these hyperparameter settings in Table 4, which are employed in our final experimental setup.

B.3 More Experimental Results

See Figures 6 ~ 15, and Tables 5 and 6.

C Assumptions

We introduce a set of standard smoothness and alignment assumptions that characterize the optimization geometry of GCL-based recommendation models during training. These assumptions are mild and are commonly adopted in nonconvex optimization and contrastive learning analysis.

Assumption A1 (Smoothness of Non-uniformity). The non-uniformity measure $\psi(\mathbf{E})$ is continuously differentiable and L_ψ -smooth with respect to the embedding matrix $\mathbf{E}$, i.e.,

$$\|\nabla\psi(\mathbf{E}_1) - \nabla\psi(\mathbf{E}_2)\|_F \leq L_\psi \|\mathbf{E}_1 - \mathbf{E}_2\|_F, \quad \forall \mathbf{E}_1, \mathbf{E}_2.$$

Assumption A2 (Gradient Alignment at Transition Start). At the beginning of the transition phase ($t = T_1$), the gradient of the ranking loss is positively aligned with the direction of increasing non-uniformity, i.e.,

$$\langle -\nabla\mathcal{L}_{\text{BPR}}(\mathbf{E}_{T_1}), \nabla\psi(\mathbf{E}_{T_1}) \rangle \geq \alpha, \quad \text{for some } \alpha > 0.$$

This assumption reflects the fact that optimizing the ranking objective encourages embedding clustering, which inherently increases non-uniformity.

Victim Model	Defense	NDCG@50↑					HR@50↓					Trade-off↑			
		NoAtt	Rand	PGA	A_ra	CLear	NoAtt	Rand	PGA	A_ra	CLear	Rand	PGA	A_ra	CLear
SimGCL	Origin	0.3757	0.3764	<u>0.3774</u>	<u>0.3770</u>	<u>0.3758</u>	0.0000	0.2086	0.0486	0.1804	0.1443	-	-	-	-
	Degree_SAD	0.3645	0.3653	0.3643	0.3642	0.3659	0.0000	0.0238	0.0155	0.0263	0.0216	0.4282	0.3229	0.4101	0.4120
	Anti-FU	0.3686	0.3709	0.3700	0.3706	0.3703	0.0000	<u>0.0190</u>	0.0165	0.0123	0.0101	0.4473	0.3202	0.4574	0.4578
	Trust-GRS	0.3573	0.3560	0.3606	0.3315	0.3599	0.0000	0.0353	0.0290	0.0435	0.0512	0.3882	0.1798	0.3189	0.3015
	w/o_p	<u>0.3759</u>	0.3778	0.3785	0.3779	0.3716	0.0000	0.0560	0.0180	<u>0.0068</u>	<u>0.0073</u>	0.3676	0.3158	0.4823	<u>0.4690</u>
	w/o_t	0.3725	<u>0.3768</u>	0.3767	0.3767	0.3706	0.0000	0.0246	<u>0.0127</u>	0.0114	0.0075	0.4415	0.3681	0.4679	0.4682
XSimGCL	GCL-Detox	0.3762	0.3715	0.3666	0.3666	0.3840	0.0000	0.0133	0.0033	0.0034	0.0061	0.4615	0.4518	0.4768	0.4821
	Origin	<u>0.3887</u>	<u>0.3882</u>	<u>0.3901</u>	<u>0.3881</u>	<u>0.3873</u>	0.0000	0.1777	0.0896	0.1304	0.2227	-	-	-	-
	Degree_SAD	0.3768	0.3767	0.3730	0.3765	0.3753	0.0000	0.0230	0.0275	0.0160	0.0250	0.4206	0.3245	0.4238	0.4283
	Anti-FU	0.3848	0.3806	0.3812	0.3801	0.3804	0.0000	0.0245	0.0187	0.0187	0.0116	0.4213	0.3840	0.4179	0.4651
	Trust-GRS	0.3729	0.3736	0.3846	0.3593	0.3669	0.0000	0.0307	0.0000	0.0233	0.0216	0.3950	0.4929	0.3734	0.4252
	w/o_p	0.3827	0.3832	0.3823	0.3825	0.3813	0.0000	0.0408	0.0235	0.0113	0.0085	0.3788	0.3591	0.4495	<u>0.4732</u>
SGL	w/o_t	0.3796	0.3799	0.3794	0.3789	0.3770	0.0000	<u>0.0178</u>	<u>0.0138</u>	<u>0.0096</u>	<u>0.0083</u>	0.4392	0.4093	0.4516	0.4682
	GCL-Detox	0.3913	0.3892	0.3907	0.3897	0.3903	0.0000	0.0106	0.0185	0.0069	0.0061	0.4714	0.3974	0.4756	0.4821
	Origin	0.3637	0.3630	0.3610	0.3622	0.3597	0.0000	0.5785	0.2929	0.4816	0.1659	-	-	-	-
	Degree_SAD	0.3349	0.3343	0.3315	0.3323	0.3331	0.0000	0.0222	0.0174	0.0181	0.0070	0.4413	0.4294	0.4399	0.4418
	Anti-FU	0.2937	0.2966	0.2922	0.2938	0.2968	0.0000	0.0355	0.0784	0.0459	0.0123	0.3779	0.2709	0.3579	0.3756
	Trust-GRS	0.2930	0.2863	0.3202	0.2874	0.2890	0.0000	0.0557	0.0531	0.0535	0.0164	0.3462	0.3529	0.3411	0.3521
SGL	w/o_p	<u>0.3630</u>	<u>0.3623</u>	0.3631	<u>0.3617</u>	<u>0.3586</u>	0.0000	0.0334	0.0303	0.1475	<u>0.0024</u>	0.4702	0.4512	0.3461	<u>0.4913</u>
	w/o_t	0.3565	0.3564	0.3567	0.3582	0.3566	0.0000	<u>0.0018</u>	0.0000	<u>0.0045</u>	0.0035	0.4894	0.4940	0.4898	0.4850
	GCL-Detox	0.3539	<u>0.3626</u>	<u>0.3628</u>	0.3586	0.3650	0.0000	0.0008	<u>0.0004</u>	0.0009	0.0009	0.4988	0.5019	0.4941	0.5045

Table 5: Performance comparison on ML-10M dataset. Best results are in bold, and second best are underlined.

Victim Model	Defense	NDCG@50↑					HR@50↓					Trade-off↑			
		NoAtt	Rand	PGA	A_ra	CLear	NoAtt	Rand	PGA	A_ra	CLear	Rand	PGA	A_ra	CLear
SimGCL	Origin	0.1403	0.1401	0.1400	0.1403	0.1399	0.0000	0.4947	0.6360	0.1027	0.6583	-	-	-	-
	Degree_SAD	0.1238	0.1219	0.1220	0.1213	0.1204	0.0000	0.0166	0.0094	0.0018	<u>0.0116</u>	0.4181	<u>0.4280</u>	0.4237	0.4214
	AntiFU	<u>0.1258</u>	0.1214	0.1206	0.1223	0.1225	0.0000	0.0122	<u>0.0075</u>	0.0316	0.0169	0.4708	0.4246	0.2821	<u>0.4248</u>
	Trust-GRS	0.1169	0.1157	0.1092	0.1242	0.1108	0.0000	0.0128	0.0114	<u>0.0009</u>	0.0123	0.4001	0.3809	<u>0.4386</u>	0.3865
	w/o_p	0.1232	0.1228	0.1238	0.1247	0.1238	0.0000	0.0129	0.0655	0.0042	0.0639	<u>0.4251</u>	0.3902	0.4241	0.3936
	w/o_t	0.1202	0.1195	0.1206	0.1205	0.1203	0.0000	<u>0.0022</u>	0.0391	0.0062	0.0344	0.4240	0.3996	0.3991	0.4039
XSimGCL	GCL-Detox	0.1244	<u>0.1238</u>	<u>0.1241</u>	<u>0.1259</u>	<u>0.1253</u>	0.0000	0.0009	0.0046	0.0000	0.0089	0.4407	0.4395	0.4487	0.4411
	Origin	0.1412	0.1407	0.1406	0.1414	0.1401	0.0000	0.3502	0.5396	0.0719	0.4896	-	-	-	-
	Degree_SAD	0.1324	0.1318	0.1321	0.1329	0.1281	0.0000	0.0092	0.0323	0.141	0.0511	0.4550	0.4400	0.3718	0.4052
	AntiFU	0.1334	0.1322	0.1319	0.1341	0.1291	0.0000	0.0086	0.0286	0.0208	<u>0.0350</u>	<u>0.4573</u>	0.4426	0.3293	<u>0.4251</u>
	Trust-GRS	0.1233	0.1229	0.1243	0.1371	0.1243	0.0000	0.0299	0.0434	0.0088	0.0549	0.3940	0.4017	<u>0.4240</u>	0.3878
	w/o_p	0.1335	0.1308	0.1312	0.1328	0.1333	0.0000	<u>0.0054</u>	0.0257	<u>0.0077</u>	0.0557	0.4571	<u>0.4426</u>	0.4158	0.4189
SGL	w/o_t	0.1281	0.1288	0.1301	0.1282	0.1282	0.0000	0.0060	<u>0.0221</u>	0.0105	0.0486	0.4490	0.4421	0.3806	0.4080
	GCL-Detox	<u>0.1393</u>	<u>0.1391</u>	<u>0.1396</u>	<u>0.1407</u>	<u>0.1396</u>	0.0000	0.0038	0.0143	0.0056	0.0327	0.4888	0.4832	0.4587	0.4649
	Origin	0.1287	0.1282	0.1289	<u>0.1284</u>	0.1289	0.0000	0.0255	0.0295	0.0177	0.0468	-	-	-	-
	Degree_SAD	0.1243	0.1215	0.1213	0.1243	0.1217	0.0000	0.0024	0.044	0.0053	0.0034	0.4268	0.3954	0.3348	0.4360
	AntiFU	0.1226	0.1232	0.1240	0.1258	0.1223	0.0000	0.0026	<u>0.0025</u>	0.0058	<u>0.0033</u>	0.4287	0.4385	0.3257	<u>0.4393</u>
	Trust-GRS	0.1215	0.1189	0.1186	0.1242	0.1185	0.0000	0.0014	0.0055	0.0000	0.0089	0.4362	0.3676	0.4839	0.3641
SGL	w/o_p	0.1197	0.1194	0.1199	0.1268	0.1195	0.0000	0.0004	0.0042	0.0000	0.0042	0.4567	0.3935	0.4938	0.4185
	w/o_t	0.1186	0.1178	0.1182	0.1188	0.1186	0.0000	0.0002	0.0073	<u>0.0004</u>	0.0057	0.4552	0.3345	0.4501	0.3985
	GCL-Detox	<u>0.1253</u>	<u>0.1251</u>	<u>0.1261</u>	0.1286	<u>0.1256</u>	0.0000	<u>0.0004</u>	0.0020	0.0000	0.0015	0.4792	0.4554	0.5010	0.4705

Table 6: Comparison on Yelp dataset. Best results are in bold, and second best are underlined.

Assumption A3 (Bounded Ranking Gradient). The gradient of the ranking loss is uniformly bounded during training:

$$\|\nabla \mathcal{L}_{\text{BPR}}(\mathbf{E})\|_F \leq G, \quad \forall \mathbf{E},$$

for some constant $G > 0$.

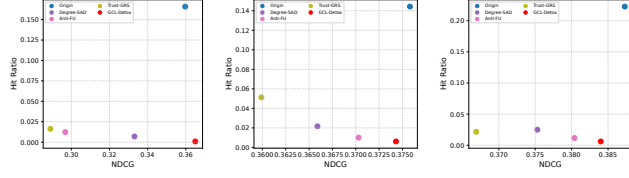
Assumption A4 (Approximate Stationarity after Contrastive Pre-training). At the end of the contrastive pretraining phase, the embedding $\mathbf{E}_{T_1}$ is an approximate stationary point of the contrastive

objective:

$$\|\nabla \mathcal{L}_{\text{CL}}(\mathbf{E}_{T_1})\|_F \leq \varepsilon,$$

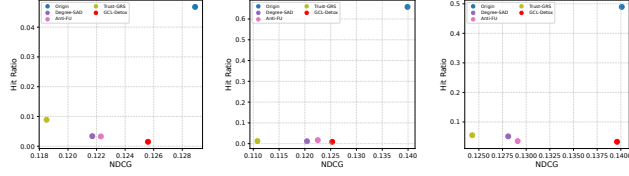
where ε is sufficiently small and its effect can be absorbed into higher-order terms in the analysis.

Remark. Assumptions (A1)–(A3) enable standard descent-lemma arguments, while (A2) captures the key geometric insight that ranking optimization naturally increases non-uniformity when starting from near-uniform embeddings. (A4) formalizes that Phase 1 provides a stable initialization. These assumptions are approximately satisfied in practice, as supported by our empirical analysis.



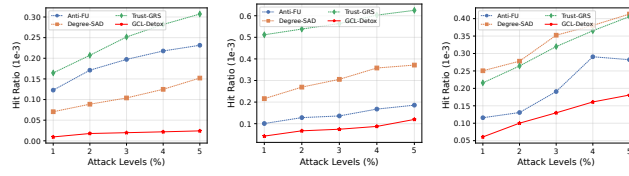
(a) SGL (b) SimGCL (c) XSimGCL

Figure 6: Pareto frontier on ML-10M with three victim models.



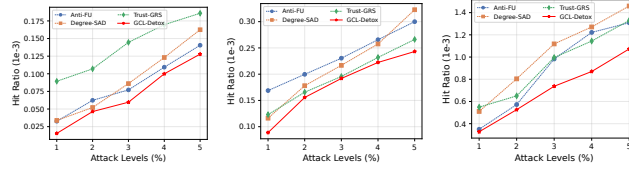
(a) SGL (b) SimGCL (c) XSimGCL

Figure 7: Pareto frontier on Yelp with three victim models.



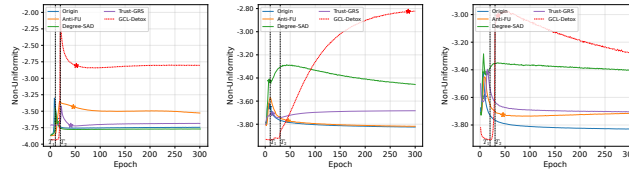
(a) SGL (b) SimGCL (c) XSimGCL

Figure 8: HR@50 under CLeaR with varying poisoning budgets (1%–5% fake users) on ML-10M.

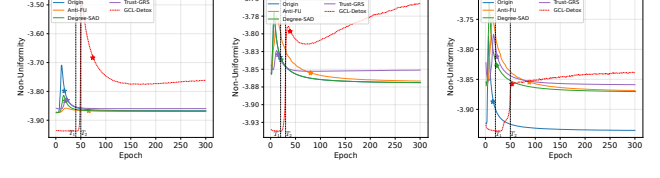


(a) SGL (b) SimGCL (c) XSimGCL

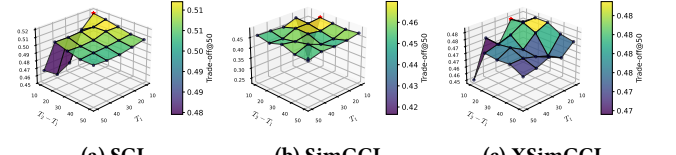
Figure 9: HR@50 under CLeaR with varying poisoning budgets (1%–5% fake users) on Yelp.



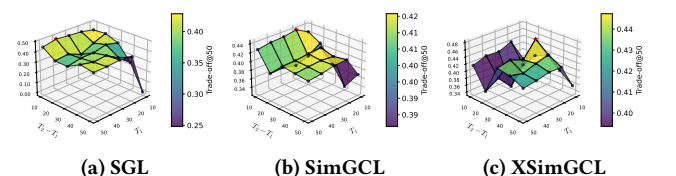
(a) SGL (b) SimGCL (c) XSimGCL

Figure 10: Evolution of the (non-)uniformity metric over epochs on ML-10M under CLeaR. T_1 and T_2 indicate the phase boundaries of GCL-Detox.

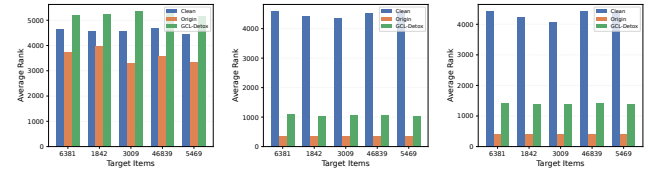
(a) SGL (b) SimGCL (c) XSimGCL

Figure 11: Evolution of the (non-)uniformity metric over epochs on Yelp under CLeaR. T_1 and T_2 indicate the phase boundaries of GCL-Detox.

(a) SGL (b) SimGCL (c) XSimGCL

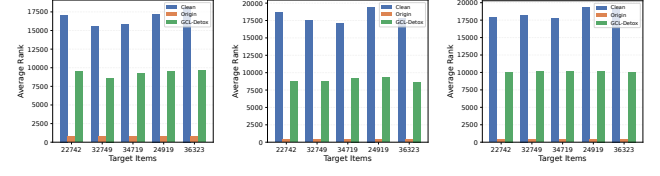
Figure 12: Trade-off@50 under CLeaR on ML-10M when tuning (T_1 , T_2) (Phase 1 end and Phase 2 end).

(a) SGL (b) SimGCL (c) XSimGCL

Figure 13: Trade-off@50 under CLeaR on Yelp when tuning (T_1 , T_2) (Phase 1 end and Phase 2 end).

(a) SGL (b) SimGCL (c) XSimGCL

Figure 14: Average ranking positions of five unpopular target items under CLeaR on ML-10M. Lower ranks indicate worse recommendation positions.



(a) SGL (b) SimGCL (c) XSimGCL

Figure 15: Average ranking positions of five unpopular target items under CLeaR on Yelp. Lower ranks indicate worse recommendation positions.